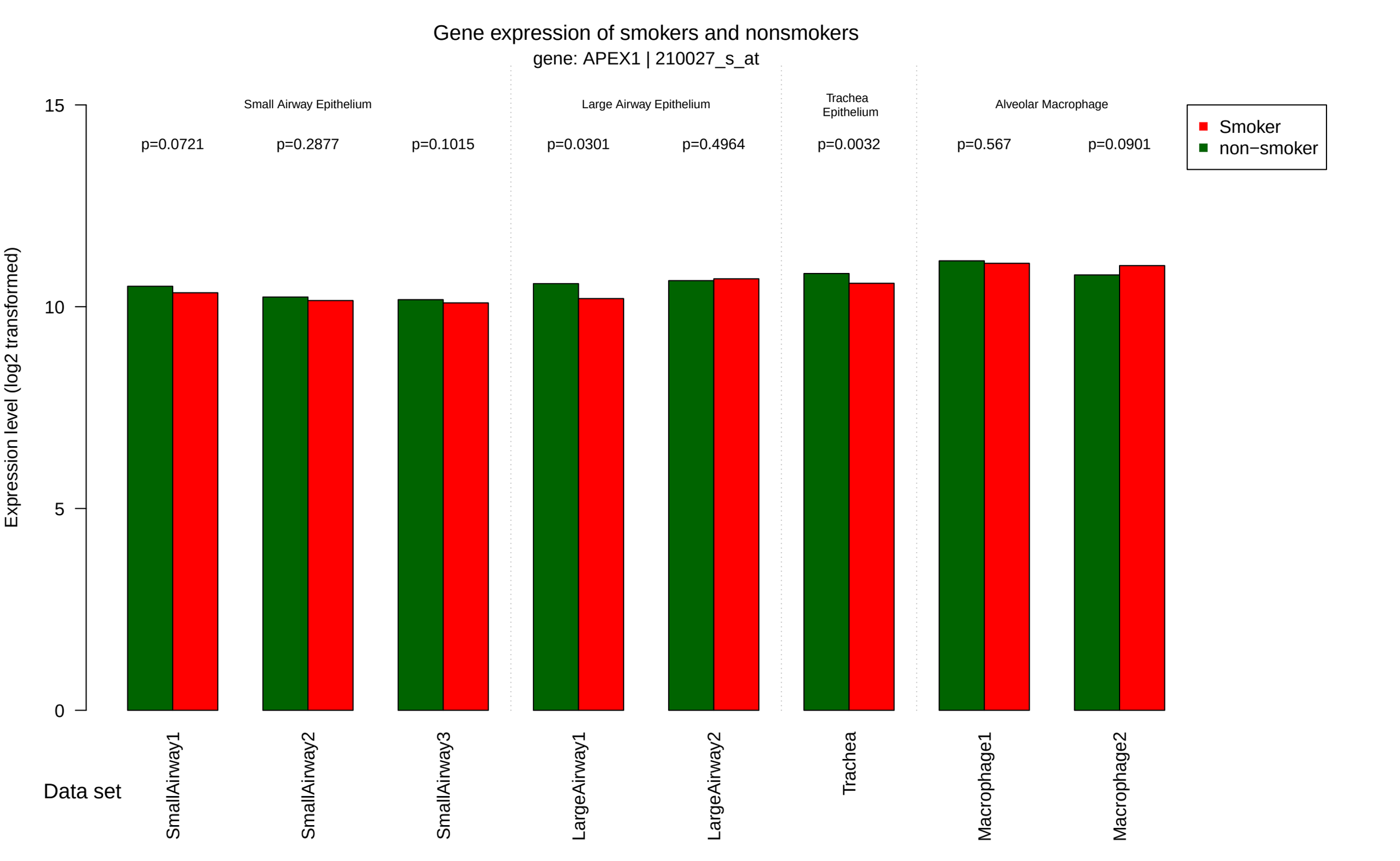
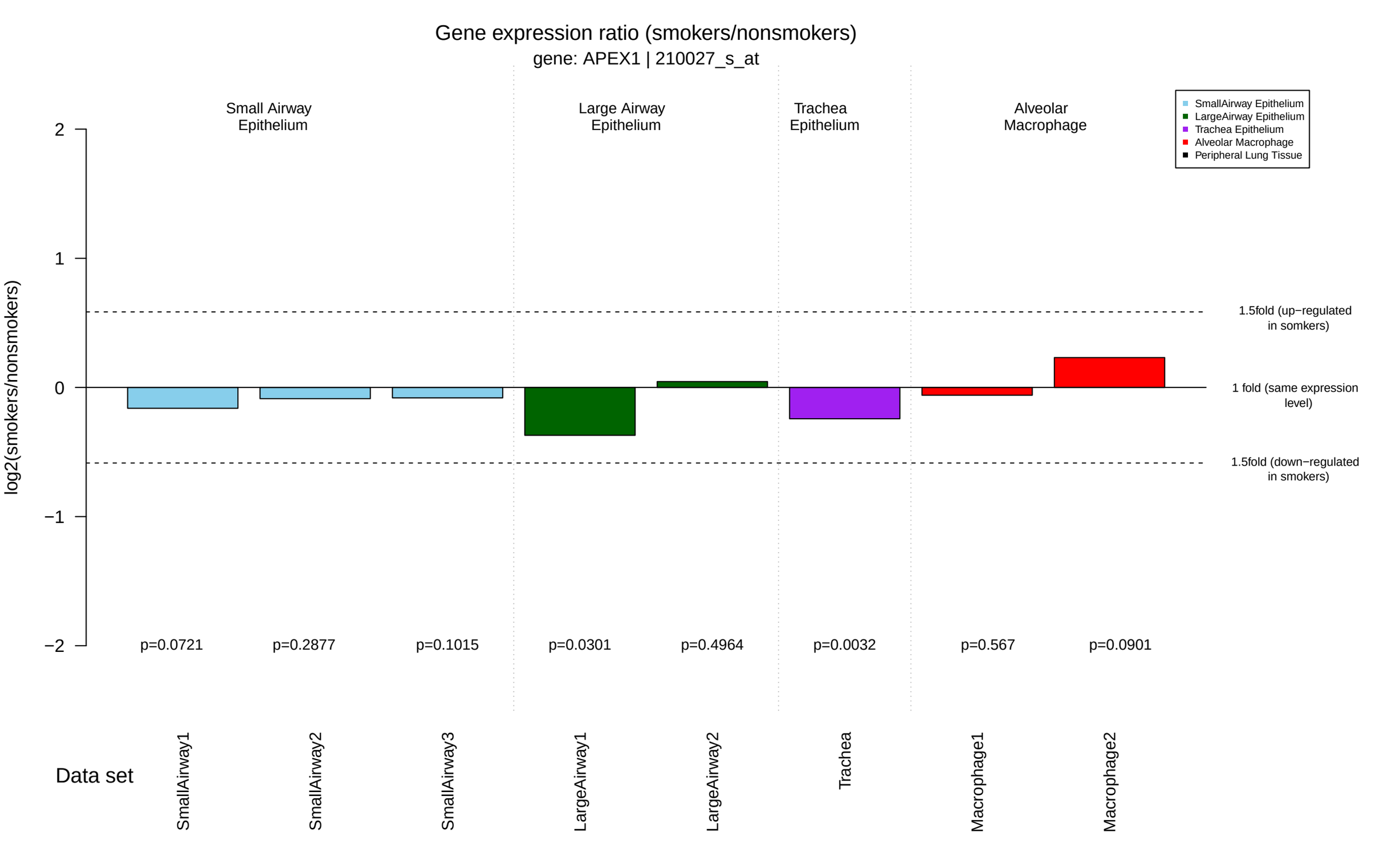
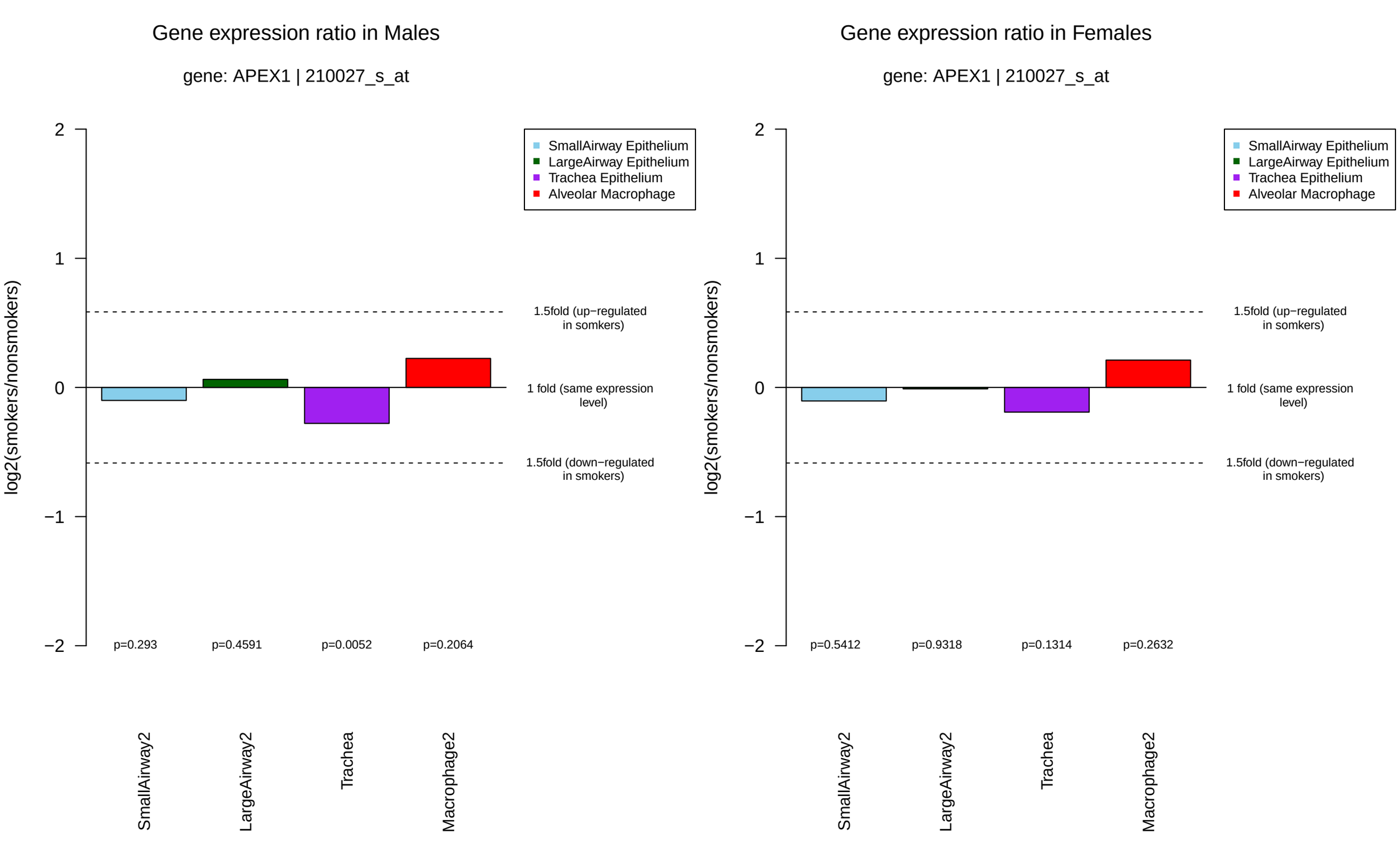
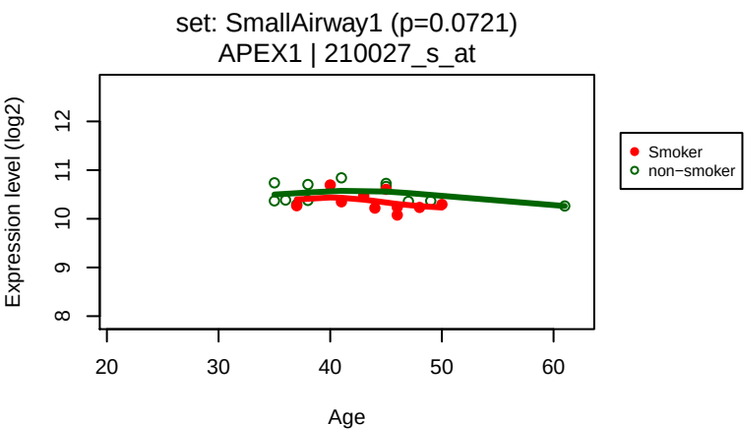
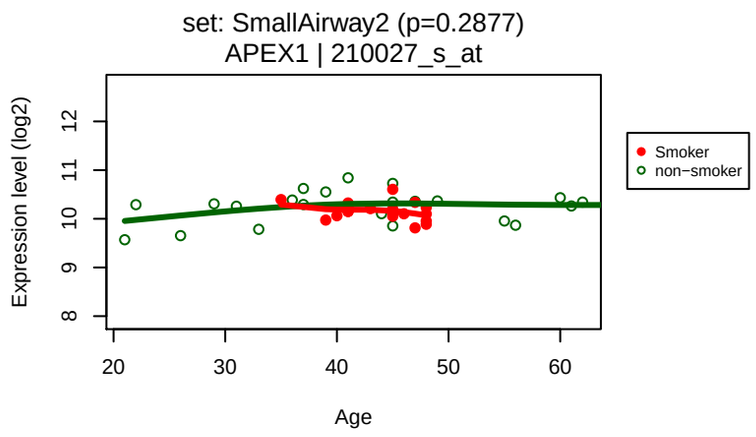
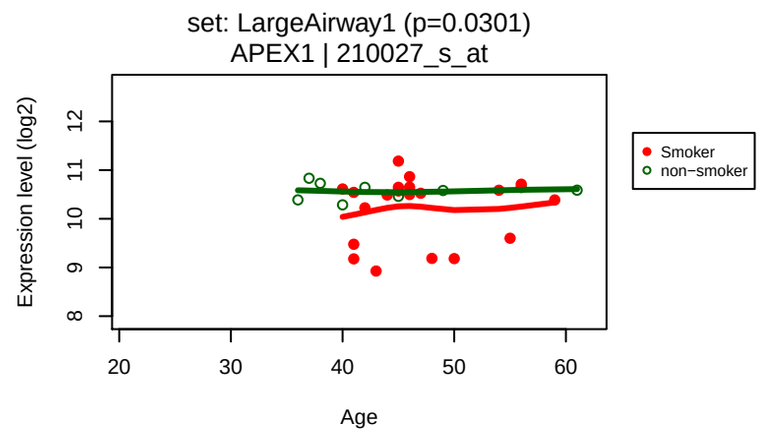

Gene expression of smokers and nonsmokers
gene: APEX1 | 210027_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0721
p=0.2877
p=0.1015
p=0.0301
p=0.4964
p=0.0032
p=0.567
p=0.0901
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

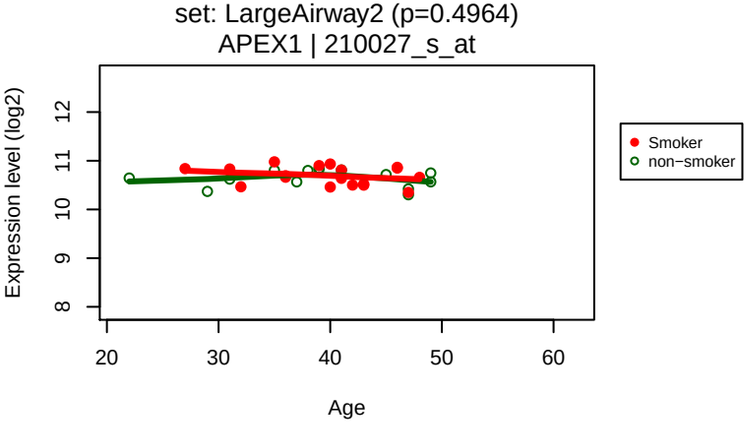

Gene expression ratio (smokers/nonsmokers)
gene: APEX1 | 210027_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0721
p=0.2877
p=0.1015
p=0.0301
p=0.4964
p=0.0032
p=0.567
p=0.0901
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

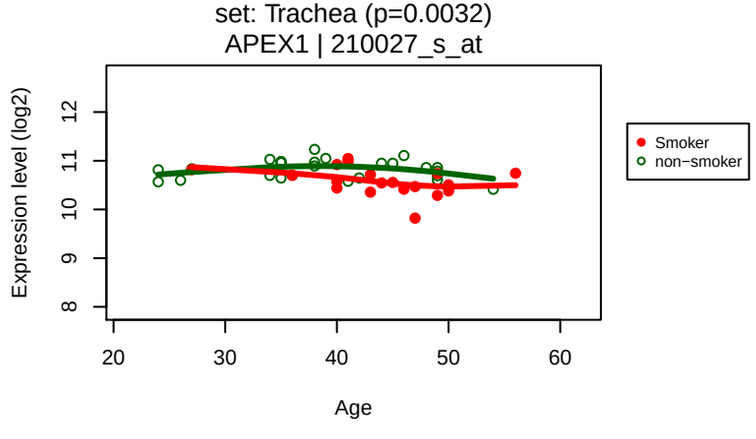

Gene expression ratio in Males
gene: APEX1 | 210027_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.293
p=0.4591
p=0.0052
p=0.2064
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: APEX1 | 210027_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5412
p=0.9318
p=0.1314
p=0.2632
SmallAirway2
LargeAirway2
Trachea
Macrophage2

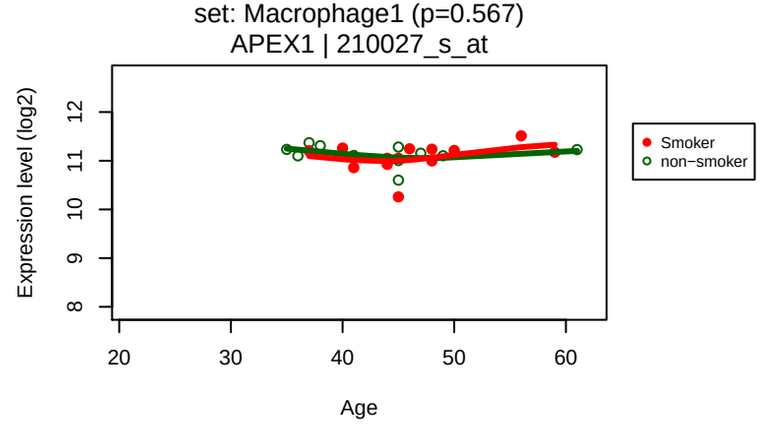

set: SmallAirway1 (p=0.0721)
APEX1 | 210027_s_at
Expression level (log2)
Age
Smoker
non−smoker

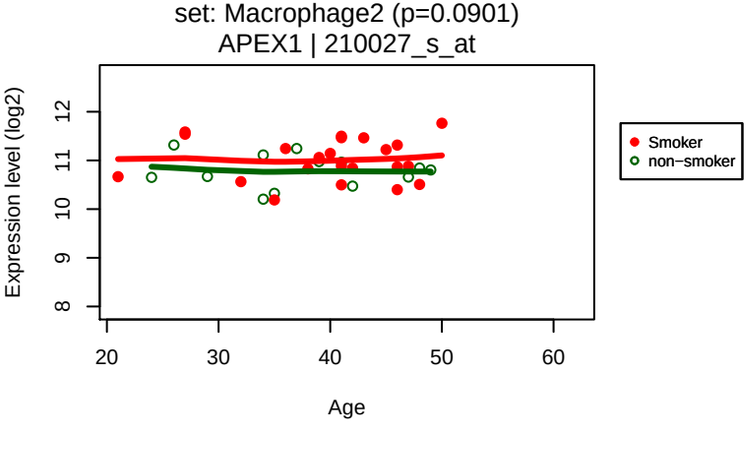

set: SmallAirway2 (p=0.2877)
APEX1 | 210027_s_at
Expression level (log2)
Age
Smoker
non−smoker

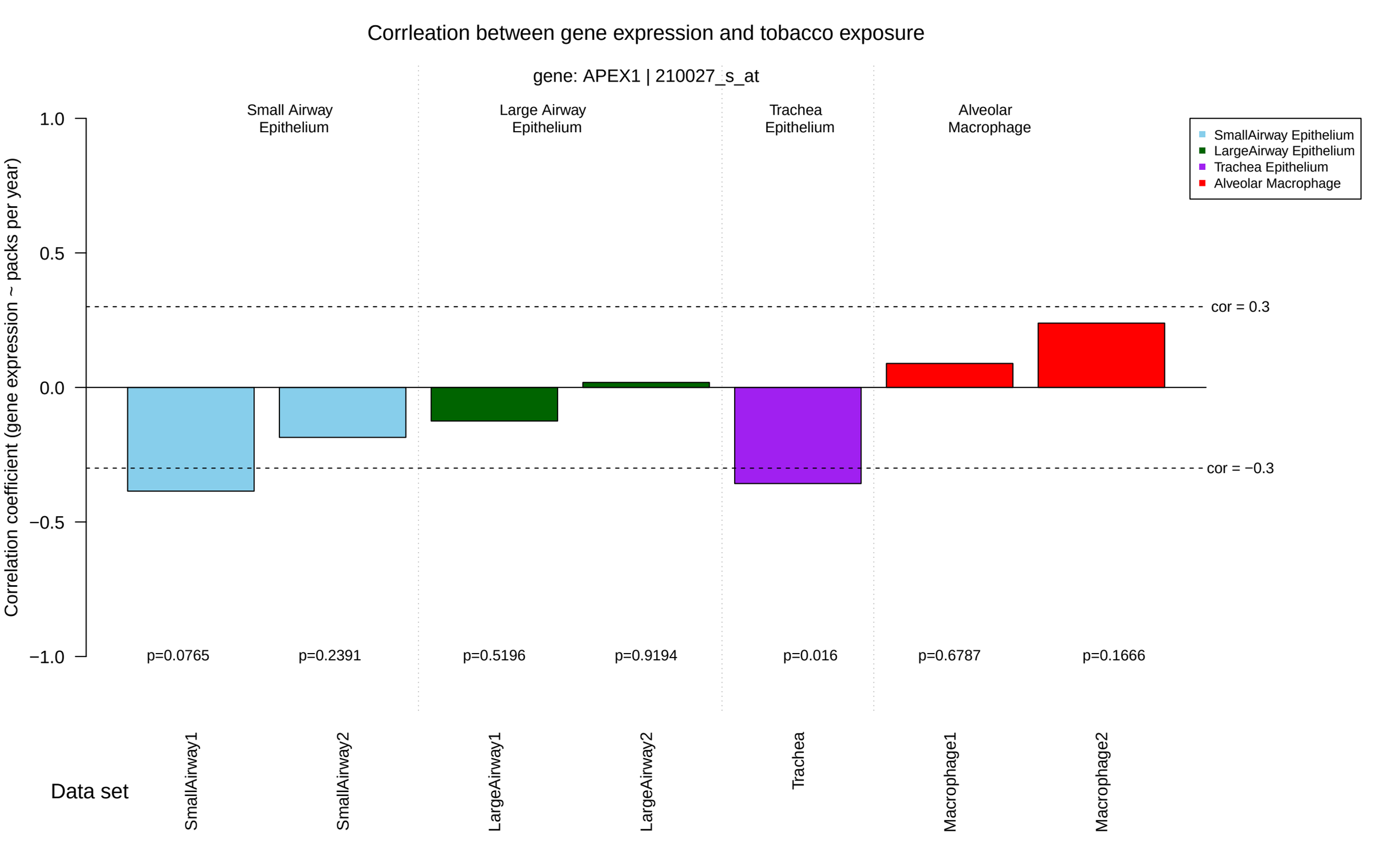

set: LargeAirway1 (p=0.0301)
APEX1 | 210027_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4964)
APEX1 | 210027_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0032)
APEX1 | 210027_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.567)
APEX1 | 210027_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0901)
APEX1 | 210027_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: APEX1 | 210027_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0765
p=0.2391
p=0.5196
p=0.9194
p=0.016
p=0.6787
p=0.1666
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2